Entanglement

If a state can be written as a product of two single-qubit states, it is called a **product state**.

Eg:
$$|\psi_{\text{prod}}\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |01\rangle)$$
$$= |0\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

Both $|0\rangle$ and $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ are single-qubit states, so $|\psi_{\text{prod}}\rangle$ is a product state.

(Note: $\otimes$ is the tensor product symbol. It is the notation used for combining multiple qubit states. Often it is omitted as a shorthand, for example $|0\rangle \otimes |0\rangle = |0\rangle|0\rangle = |00\rangle$. All of these states are equivalent and may be used interchangeably.)

A state that CANNOT be written as a product of two single-qubit states is called an **entangled state**.

For a two-qubit system there are four important entangled states known as the **Bell states**.

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \quad |\Psi^-\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$$
$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) \quad |\Phi^-\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

Entanglement is important because it can be used as a resource to speed up computations with a quantum computer.

Measurement on multiple qubit states

The general state for a 2 bit state is :

$$|\psi\rangle = a|0\rangle_1|0\rangle_0 + b|0\rangle_1|1\rangle_0 + c|1\rangle_1|0\rangle_0 + d|1\rangle_1|1\rangle_0$$

Example: What is the probability of finding the 0th qubit in the state $|0\rangle$?

We want to know $\|\langle 0|_0|\psi\rangle\|^2$. We will calculate $\langle 0|_0|\psi\rangle$ first, then find its modulus squared.

$$\begin{aligned}\langle 0|_0|\psi\rangle &= \langle 0|_0(a|0\rangle_1|0\rangle_0 + b|0\rangle_1|1\rangle_0 + c|1\rangle_1|0\rangle_0 + d|1\rangle_1|1\rangle_0) \\ &= a|0\rangle_1\langle 0|_0|0\rangle_0 + b|0\rangle_1\langle 0|_0|1\rangle_0 \\ &+ c|1\rangle_1\langle 0|_0|0\rangle_0 + d|1\rangle_1\langle 0|_0|1\rangle_0)\end{aligned}$$

Recall that the inner product of two quantum states is 1 if they are the same state and is 0 if they are orthogonal states.

$$\langle 0|_0|\psi\rangle = a|0\rangle_1 + c|1\rangle_1$$

Final answer: $\|a|0\rangle_1 + c|1\rangle_1\|^2 = \|a + c\|^2$

After we make a measurement, the state collapses to only the terms with the 0th qubit in the $|0\rangle$ state. Here: $a|0\rangle_1 + c|1\rangle_1$
However, the coefficients a and c now do not correctly represent measurement outcomes, $|a|^2 + |c|^2$ does not necessarily = 1.
To fix this we divide each coefficient by the probability we have just worked out: $\|a + c\|^2$

State after measurement (SAM):

$$|\psi_{SAM}\rangle = \frac{1}{\|a + c\|^2}(a|0\rangle_1 + c|1\rangle_1)$$

Teleportation

GOAL: Transfer a secret quantum state from Alice to Bob, while they are separated from each other.

STEP ONE: Alice and Bob entangle a pair of qubits.
Initial state: $|0\rangle_B|0\rangle_A$
Apply H gate on qubit A, then CNOT with qubit A as control and qubit B as target.

Final state: $\frac{1}{\sqrt{2}}(|0\rangle_B|0\rangle_A + |1\rangle_B|1\rangle_A)$
They can now separate their states, but they will remain entangled

STEP TWO: Alice prepares a secret state on qubit Q.
Secret state: $|\psi\rangle_Q = \alpha_0|0\rangle_Q + \alpha_1|1\rangle_Q$
This is the state we will be transferring to Bob. Note: we can't measure this state to tell Bob which state he needs, because it will simply output $|0\rangle$ or $|1\rangle$.

Final state: $\frac{1}{\sqrt{2}}(|0\rangle_B|0\rangle_A + |1\rangle_B|1\rangle_A)(\alpha_0|0\rangle_Q + \alpha_1|1\rangle_Q)$

STEP THREE: Alice entangles A with Q.
Initial state: $\frac{1}{\sqrt{2}}(|0\rangle_B|0\rangle_A + |1\rangle_B|1\rangle_A)(\alpha_0|0\rangle_Q + \alpha_1|1\rangle_Q)$
Apply H gate on qubit A, then CNOT with qubit A as control and qubit Q as target.

Final state: $$\begin{aligned}\psi\rangle = \frac{1}{2}&(\alpha_0|0\rangle_B + \alpha_1|1\rangle_B)|0\rangle_A|0\rangle_Q + (\alpha_0|0\rangle_B - \alpha_1|1\rangle_B)|0\rangle_A|1\rangle_Q \\ &+ (\alpha_1|0\rangle_B + \alpha_0|1\rangle_B)|1\rangle_A|0\rangle_Q + (-\alpha_1|0\rangle_B + \alpha_0|1\rangle_B)|1\rangle_A|1\rangle_Q)\end{aligned}$$

Since both B and Q are entangled with A, acting on Q can impact B even if the two states never meet.

Teleportation

STEP FOUR: Alice measures qubits A and Q, then communicates her result to Bob classically (using 2 bits of classical information).
This tells Bob what to do to his state to retrieve the secret state Q.

Alice outcome	Bob's state	Instruction to Bob	Result
$\|0\rangle_A\|0\rangle_Q$	$\alpha_0\|0\rangle_B + \alpha_1\|1\rangle_B$	None	$\alpha_0\|0\rangle_B + \alpha_1\|1\rangle_B$
$\|0\rangle_A\|1\rangle_Q$	$\alpha_0\|0\rangle_B - \alpha_1\|1\rangle_B$	Z	$\alpha_0\|0\rangle_B + \alpha_1\|1\rangle_B$
$\|1\rangle_A\|0\rangle_Q$	$\alpha_1\|0\rangle_B + \alpha_0\|1\rangle_B$	X	$\alpha_0\|0\rangle_B + \alpha_1\|1\rangle_B$
$\|1\rangle_A\|1\rangle_Q$	$-\alpha_1\|0\rangle_B + \alpha_0\|1\rangle_B$	X then Z	$\alpha_0\|0\rangle_B + \alpha_1\|1\rangle_B$